CS257 Linear and Convex Optimization
Homework 12

Due: December 14, 2020

December 7, 2020

1. Consider the following problem,

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^2} \quad & f(\boldsymbol{x}) = x_1^2 + x_2^2 \\ \text{s.t.} \quad & g_1(\boldsymbol{x}) = (x_1 - 1)^2 + (x_2 - 1)^2 - 1 \leq 0 \\ & g_2(\boldsymbol{x}) = (x_1 - 1)^2 + (x_2 + \frac{1}{2})^2 - 1 \leq 0 \end{aligned}$$

Write down the KKT conditions and find the optimal point $\boldsymbol{x}^*$ and the corresponding Lagrange multipliers.

Solution. The KKT conditions are

$$\begin{cases} \mu_1 \geq 0 \\ \mu_2 \geq 0 \\ 2x_1 + 2\mu_1(x_1 - 1) + 2\mu_2(x_1 - 1) = 0 \\ 2x_2 + 2\mu_1(x_2 - 1) + 2\mu_2(x_2 + \frac{1}{2}) = 0 \\ \mu_1[(x_1 - 1)^2 + (x_2 - 1)^2 - 1] = 0 \\ \mu_2[(x_1 - 1)^2 + (x_2 + \frac{1}{2})^2 - 1] = 0 \\ (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1 \\ (x_1 - 1)^2 + (x_2 + \frac{1}{2})^2 \leq 1 \end{cases}$$

Case I. Both g_1 and g_2 are active. Then

$$\begin{cases} (x_1 - 1)^2 + (x_2 - 1)^2 = 1 \\ (x_1 - 1)^2 + (x_2 + \frac{1}{2})^2 = 1 \end{cases}$$

so

$$x_2 = \frac{1}{4}, \quad x_1 = 1 \pm \frac{\sqrt{7}}{4}$$

Apparently $(1+\frac{\sqrt{7}}{4}, \frac{1}{4})^T$ cannot be the optimal point. Plugging $(1-\frac{\sqrt{7}}{4}, \frac{1}{4})^T$ into the stationarity condition,

$$\begin{cases} 2-\frac{\sqrt{7}}{2}-\frac{\sqrt{7}}{2}\mu_1-\frac{\sqrt{7}}{2}\mu_2=0 \\ \frac{1}{2}-\frac{3}{2}\mu_1+\frac{3}{2}\mu_2=0 \end{cases} \implies \begin{cases} \mu_1=\frac{2}{\sqrt{7}}-\frac{1}{3}>0 \\ \mu_2=\frac{2}{\sqrt{7}}-\frac{2}{3}>0 \end{cases}$$

Since the objective function is strictly convex, $\boldsymbol{x}^*=(1-\frac{\sqrt{7}}{4}, \frac{1}{4})^T$ is the unique global minimum. But let's check the other cases as well

Case II. g_1 is active, g_2 is inactive. Then $\mu_2=0$. Plugging into the stationarity condition,

$$\begin{cases} 2x_1+2\mu_1(x_1-1)=0 \implies x_1=\frac{\mu_1}{1+\mu_1} \\ 2x_2+2\mu_1(x_2-1)=0 \implies x_2=\frac{\mu_1}{1+\mu_1} \end{cases}$$

In particular $x_1=x_2\in[0,1)$. Using $g_1(\boldsymbol{x})=0$,

$$x_1=x_2=1-\frac{\sqrt{2}}{2}.$$

But it violates $g_2(\boldsymbol{x})<0$, as

$$(x_1-1)^2+(x_2+\frac{1}{2})^2=\frac{13-6\sqrt{2}}{4}>1$$

Case III. g_2 is active, g_1 is inactive. Then $\mu_1=0$. Plugging into the stationarity condition,

$$\begin{cases} 2x_1+2\mu_2(x_1-1)=0 \implies x_1=\frac{\mu_2}{1+\mu_2} \\ 2x_2+2\mu_2(x_2+\frac{1}{2})=0 \implies x_2=-\frac{\mu_2}{2+2\mu_2} \end{cases}$$

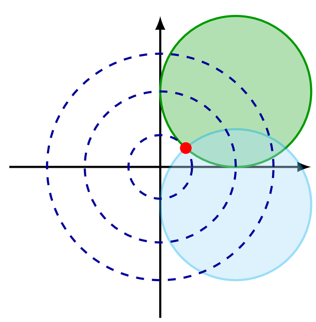
Plugging into $g_2(\boldsymbol{x})=0$,

$$\mu_2=\frac{\sqrt{5}}{2}-1, \quad x_1=1-\frac{2}{\sqrt{5}}, \quad x_2=-\frac{1}{2}+\frac{1}{\sqrt{5}}.$$

But it violates $g_1(\boldsymbol{x})<0$, as

$$(x_1-1)^2+(x_2-1)^2=\frac{13}{4}-\frac{3}{\sqrt{5}}>1$$

Case IV. Both g_1 and g_2 are inactive. Then $\mu_1=\mu_2=0$. Plugging into the stationarity condition, we obtain $x_1=x_2=0$, violating both $g_1(\boldsymbol{x})<0$ and $g_2(\boldsymbol{x})<0$.



$\square$

2. Consider the following problem,

$$\begin{aligned}
\min_{\boldsymbol{x}\in\mathbb{R}^2} \quad & f(\boldsymbol{x}) = (x_1-1)^2 + (x_2+1)^2 \\
\text{s.t.} \quad & g(\boldsymbol{x}) = x_1 - x_2 + 1 \le 0
\end{aligned}$$

(a). Write down the KKT conditions and find the optimal point, the corresponding Lagrange multiplier and the optimal value.

(b). Find the Lagrange dual function. Note that we do not require $\mu \ge 0$ in the dual function.

(c). Derive the Lagrange dual problem.

(d). Write down the KKT conditions for the Lagrange dual problem. Find the optimal point, the corresponding Lagrange multiplier and the optimal value.

Solution. (a). The KKT conditions are

$$\begin{cases}
\mu \ge 0 \\
2(x_1-1)+\mu = 0 \\
2(x_2+1)-\mu = 0 \\
\mu(x_1-x_2+1) = 0 \\
x_1 - x_2 + 1 \le 0
\end{cases}$$

The optimal solution $x_1 = -\frac{1}{2}$, $x_2 = \frac{1}{2}$, $\mu = 3$. The optimal value $f^* = \frac{9}{2}$.

(b). The Lagrange dual function is

$$\begin{aligned}
\phi(\mu) &= \inf_{\boldsymbol{x}\in\mathbb{R}^2} [(x_1-1)^2 + (x_2+1)^2 + \mu(x_1-x_2+1)] \\
&= \inf_{x_1\in\mathbb{R}} (x_1^2 - 2x_1 + \mu x_1) + \inf_{x_2\in\mathbb{R}} (x_2^2 + 2x_2 - \mu x_2) + \mu + 2 \\
&= -\frac{1}{4}(\mu^2 - 4\mu + 4) - \frac{1}{4}(\mu^2 - 4\mu + 4) + \mu + 2 \\
&= -\frac{1}{2}\mu^2 + 3\mu, \quad \mu \in \mathbb{R}
\end{aligned}$$

(c). The Lagrange dual problem is

$$\begin{aligned}
\max_{\mu} \quad & -\frac{1}{2}\mu^2 + 3\mu \\
\text{s.t.} \quad & \mu \ge 0
\end{aligned}$$

(d). The KKT conditions for the dual problem are

$$\begin{cases}
\mu' \ge 0 \\
-\mu + 3 + \mu' = 0 \\
\mu'\mu = 0 \\
\mu \ge 0
\end{cases}$$

The optimal solution $\mu = 3$, the Lagrange multiplier $\mu' = 0$. The optimal value $f^* = \frac{9}{2}$.

$\square$